

2 million IE9 downloads

Microsoft recorded over 2 million downloads of the new Internet Explorer 9 Beta

**Google white space**

Google is currently testing broadband using white space spectrum in Ohio, USA; white space are unused radio frequency ranges

HP incubating 'memristors' with Hynix

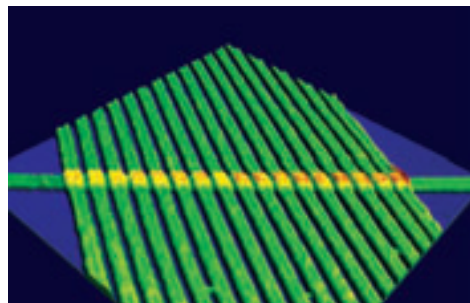
If you are familiar with any of the terms resistance, capacitance or inductance, you will know that there are three properties which an electric circuit is bound to have. In the late 1960s, a prediction was made for yet another circuitual property, then dubbed as Memristance. IEEE confirmed the property renaming to memristance (and the circuitual element showing this property is called memristor) in 1971.

Though it is not a grand revelation, the mem of the term memristor comes from memory. This property allows a circuit element to memorise its state – even after it is switched off and then switched back on. This made research labs try and work out the feasibility to use it as a memory solution. The good news is, of late – the fruit is about to ripen. Scientists at HP Labs have confirmed the possibility to make storage chips based on memristors for sometime now, and now they have taken a real big step forward, collaborating with Hynix to produce RAM for the new generation.

Hynix is an old player in memory solutions, and HP found them good to work with to take the technology further, and to show us a glimpse of the new era. The new system memory solutions are primarily dubbed as Resistive Random Access Memory (ReRAM).

The possibilities of memristors are endless, as HP explains it. Even though they are open to work with other companies (yet

to be announced) other than Hynix, HP says they'll keep the intellectual property to themselves, and make it a licensable standard to be used by other chip manufacturers.



We will have to see how interesting the race between flash and memristor memories gets, but in the race of S/DRAMs and ReRAM, the legacy will fall flat without a doubt. As you already know, no matter how speedy it gets, once switched off, a DDR module is as good as a breadboard. On the other hand, a ReRAM module will be completely able to retain its state even after the PC is turned off. Hang on, is that supposed to mean we'll have faster boot-up? Yes, a lot faster – a restart will be much like Windows Sleep Mode.

Let's see where HP-Hynix leads us; the future is promising enough to be brighter – but the question is, how much? 

AMD showcases Bulldozer and Bobcat

AMD showed off its next-gen processor technology at the Hot Chips 2010 conference in Silicon Valley, detailing the evolved nature of its upcoming Bulldozer and Bobcat lineup of chips.

"The new cores, codenamed 'Bulldozer' for high-performance PC and server markets, and 'Bobcat' for low-power notebook and small form-factor desktop markets, were designed from the ground-up to address specific requirements and compute workloads," said Chekib Akrouit, a senior VP at AMD.

AMD has apparently taken its x86 processor technology to the next level, radically evolving its core designs from



the first time since the original Opteron x86 and K7 processors. To highlight this new ground-up approach, Akrouit went on to say: "x86 architecture lies at the very heart of computing and AMD has continuously evolved and improved its core designs. The Bulldozer and Bobcat cores continue that evolutionary path and are designed to change the user's experience with the resulting products."

One of the biggest features of the new Bulldozer architec-

ture is that it will support a new sort of SMT (simultaneous multithreading) and not the traditional CMP (chip multiprocessing). This will finally allow two or more threads to run on a single core, working simultaneously.

How this technology compares to Intel's Hyper-threading is not yet known, though it has been noted that with AMD's novel SMT approach, two threads will share a single front-end, but, use separate integer execution resources.

According to Ars Technica's Jon Stokes, for this type of SMT to work, all shared code and data storage components of the processor must be "either replicated or partitioned", giving equal priority to each thread when sharing resources, in

such a way that buffer entries don't remain empty when there's no second thread.

Not many definitive statements can be made about the Bulldozer family of processors, but you can safely expect some rather fantastic floating-point performance, especially if there's sufficient memory bandwidth.

Meanwhile, the Bobcat lineup, while great for laptops and tablets, might just be AMD's best bet to take on ARM's Eagle lineup in the high-density cloud server market, the wave of the future.

It really does seem as if AMD has all future bases covered, from desktop, server, and laptop to tablet markets, especially if you include the Fusion Accelerated Process Unit (APU). 